

JEE MAIN REVISION MASTER: MATHEMATICAL REASONING 2026 1 QUESTIONS 1.67% WEIGHTAGE

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 9 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Moderate	Mathematical Reasoning — Standard Calculations & Problem Archetypes	6 PYQs	Quantitative Analysis
Standard	Mathematical Reasoning — Core Principles & Governing Laws	2 PYQs	Core Principle
Standard	Mathematical Reasoning — Applied Cases, Reactions & Traps	2 PYQs	Applied Topics

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Mathematical Reasoning — Standard Calculations & Problem Archetypes	$n=4, \quad P(n):2^{2n}-1, \quad 3^{\{ \}}$
Mathematical Reasoning — Core Principles & Governing Laws	$3^{\{ \}}$
Mathematical Reasoning — Applied Cases, Reactions & Traps	$x;x^{2}+5$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Mathematical Reasoning — Standard Calculations & Problem Archetypes (Moderate Yield — 6 PYQs)

Core Principle: Quantitative numerical problem-solving techniques and standard question patterns frequently seen in Mathematical Reasoning.

■■ **Caution / JEE Pitfall:** Watch for algebraic rounding, stoichiometric balancing, and boundary condition checks.

■ **Speed Trick:** Dimensional consistency checks and ratio comparisons can quickly eliminate incorrect options.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "The solution of $3(x-1) \leq 2(x-3)$ is..." [Ans: A]
- [2020 Shift] Q#1 (MCQ): "The negation of the statement "For all real numbers x and y , $x+y=y+x$ " is..." [Ans: N/A]
- [2020 Shift] Q#4 (MCQ): "If $P(n):2^n < n!$ Then the smallest positive integer for which $P(n)$ is true if..." [Ans: N/A]

3.2 Mathematical Reasoning — Core Principles & Governing Laws (Standard Yield — 2 PYQs)

Core Principle: Fundamental definitions, core postulates, and governing physical/chemical relations in Mathematical Reasoning.

■■ **Caution / JEE Pitfall:** Carefully verify units, standard reference states, and sign conventions in Mathematical Reasoning.

■ **Speed Trick:** Focus on high-yield formulas and standard limiting approximations tested in Mathematical Reasoning PYQs.

Representative Source PYQs:

- [2023 Shift] Q#3 (MCQ): "The contrapositive of the statement. "If two lines do not intersect in the same plane, then they are parallel." is..." [Ans: N/A]
- [2018 Shift] Q#4 (MCQ): "The negation of the statement "72 is divisible by 2 and 3" is..." [Ans: N/A]

3.3 Mathematical Reasoning — Applied Cases, Reactions & Traps (Standard Yield — 2 PYQs)

Core Principle: Special cases, experimental observations, and conceptual traps designed to test deep understanding of Mathematical Reasoning.

■■ **Caution / JEE Pitfall:** Read questions carefully for keywords like 'incorrect', 'not true', or non-standard conditions.

■ **Speed Trick:** Elimination of extreme or unphysical answer choices improves accuracy in assertion-reason and multiple-statement questions.

Representative Source PYQs:

- [2024 Shift] Q#2 (MCQ): "The negation of the statement "For every real number x , x^2+5 is positive" is..." [Ans: N/A]
- [2019 Shift] Q#2 (MCQ): "The negative of the statement "All continuous functions are differentiable."..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

The solution of $3(x-1) \leq 2(x-3)$ is

- (A) $x \leq -3$ [Correct]
- (B) $x \geq -3$
- (C) $x \leq 3$
- (D) $x \geq 3$

Correct Answer: A

Solution Analysis: Given inequality: $3(x-1) \leq 2(x-3)$ Open the brackets: $3x-3 \leq 2x-6$ Now subtract $2x$ from both sides: $x-3 \leq -6$ Add 3 to both sides: $x \leq -3$ So, the solution is: $x \leq -3$ Hence, the correct option is Option A .

Benchmark Problem 2: KCET 2024 (Q#2)

The negation of the statement "For every real number x ; x^2+5 is positive" is

- (A) For every real number x ; x^2+5 is not positive.
- (B) For every real number x ; x^2+5 is negative.
- (C) There exists atleast one real number x , such that x^2+5 is not positive.
- (D) There exists atleast one real number x , such that x^2+5 is positive.

Correct Answer: None

Solution Analysis: The negation of statement "For every real number x ; x^2+5 is positive" is there exists at least one real number x such that x^2+5 is not positive.

Benchmark Problem 3: KCET 2023 (Q#3)

The contrapositive of the statement. "If two lines do not intersect in the same plane, then they are parallel." is

- (A) If two lines are parallel, then they intersect in the same plane.
- (B) If two lines are not parallel, then they do not intersect in the same plane.
- (C) If two lines are parallel, then they do not intersect in the same plane.
- (D) If two lines are not parallel, then they intersect in the same plane.

Correct Answer: None

Solution Analysis: Given, statement "If two lines do not intersect in the same plane, then they are parallel. Contrapositive It two lines intersect in the same plane, then they are not parallel. Converse If two lines do not intersect in the same plane, then they are parallel.